

E-COMMERCE SALES ANALYSIS

2023 – 2025

SQL-Based Data Analytics Portfolio Project

All tables and metrics in this report were produced using SQL queries against the e-commerce transactions dataset.

1. Key Metrics

Overall performance summary across the full analyzed period (October 2023 – October 2025).

Metric	Value
Total Revenue	533,666,024
Total Profit	79,708,735
Total Orders	5,000
Total Customers	4,844
Average Order Value (AOV)	106,733
Average Margin	15%

Insights & Comments

The overall numbers point to a stable, moderately profitable business built on relatively large basket sizes rather than sheer transaction volume. A 15% margin combined with an AOV above 100,000 suggests the business is self-funding: profit generated is sufficient to sustain current operations without relying on external capital.

Given the AOV is already high, a natural next step would be to test cross-sell or bundling strategies to lift basket size further, rather than focusing purely on acquiring more customers.

2. Monthly Sales Dynamics

Revenue, profit and order volume trends on a month-by-month basis.

Month	Revenue	Profit	Orders
2023-10	21,307,522	3,106,231	202
2023-11	22,040,269	3,387,740	204
2023-12	20,624,240	2,989,331	195
2024-01	21,477,241	3,104,254	222
2024-02	19,853,397	2,941,449	196
2024-03	21,571,015	3,235,675	199
2024-04	22,214,781	3,241,580	214
2024-05	24,744,787	3,692,959	217
2024-06	21,368,963	3,261,199	201
2024-07	24,019,284	3,486,946	220
2024-08	23,299,895	3,404,785	230
2024-09	22,480,182	3,292,918	208
2024-10	23,607,627	3,601,282	218
2024-11	22,036,248	3,286,992	188

Month	Revenue	Profit	Orders
2024-12	24,806,786	3,855,730	219
2025-01	21,520,802	3,255,994	220
2025-02	19,902,446	2,970,431	178
2025-03	22,602,989	3,533,884	229
2025-04	21,653,818	3,150,564	200
2025-05	26,010,929	3,979,357	234
2025-06	21,155,496	3,184,539	203
2025-07	22,526,568	3,439,384	187
2025-08	23,317,916	3,456,953	213
2025-09	18,131,497	2,677,364	185
2025-10*	1,391,328	171,194	18

* October 2025 is a partial month.

Insights & Comments

Monthly revenue is consistently stable across the full period, generally holding in the 20–25M range with no sharp structural breaks. A handful of isolated dips are visible (e.g., Feb 2025, Jul 2025, Sep 2025), but they fall within normal month-to-month variance rather than signaling a trend reversal.

Explaining these dips with confidence would require additional context beyond this dataset (seasonality, marketing activity, external market factors), so they are flagged here as points of interest rather than conclusions.

3. Revenue, Profit, Orders and Margin by Category

Category	Revenue	Profit	Orders	Customers	AOV	Margin
Beauty	50,803,410	7,213,436	460	460	110,442	14%
Books	54,932,643	8,076,273	528	524	104,039	15%
Clothing	55,053,908	8,445,750	511	510	107,738	15%
Electronics	52,587,884	8,042,134	472	472	111,415	15%
Furniture	56,647,188	8,693,087	527	525	107,490	15%
Groceries	47,883,103	7,075,333	470	468	101,879	15%
Home Decor	57,233,222	8,556,847	515	513	111,132	15%
Kitchen	54,227,902	7,879,573	528	527	102,704	15%
Sports	52,069,397	7,739,430	511	509	101,897	15%
Toys	52,227,366	7,986,871	478	477	109,262	15%

Insights & Comments

Kitchen and Books lead in order volume, pointing to stronger demand and a broader customer base for these categories, even though their revenue is not the highest overall.

Furniture and Home Decor lead in total profit. Furniture's result is driven mainly by strong order volume and revenue rather than an exceptionally high AOV — at 107,490 its AOV is close to the overall average (106,733). Home Decor and Electronics, by contrast, carry the highest AOV among all categories.

Beauty stands out as the weakest category: it has the fewest orders and customers, and the only margin one point below the rest (14% vs. 15% elsewhere). This gap is large enough relative to the average to be worth a closer look — for example, whether pricing or cost structure differs for this category.

4. Top-5 Categories by Revenue and Profit

Category	Revenue	Profit	Orders	Customers	AOV	Margin
Furniture	56,647,188	8,693,087	527	525	107,490	15%
Home Decor	57,233,222	8,556,847	515	513	111,132	15%
Clothing	55,053,908	8,445,750	511	510	107,738	15%
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Electronics	52,587,884	8,042,134	472	472	111,415	15%

Insights & Comments

The top-5 pattern largely confirms the category-level picture above. One point worth highlighting: Electronics sustains strong profit despite not leading in order volume, driven by the highest AOV among all categories — a reminder that revenue leadership and profit leadership don't always come from the same driver (volume vs. basket size).

5. Regional Breakdown

Revenue, average check and margin by region.

Region	Revenue	Profit	Orders	Customers	AOV	Margin
North	143,578,246	21,343,004	1,288	1,283	111,474	15%
East	135,811,638	20,532,558	1,256	1,249	108,130	15%
West	131,045,973	19,580,123	1,241	1,232	105,597	15%
South	123,230,167	18,253,049	1,215	1,206	101,424	15%

Insights & Comments

North is the strongest region across every metric — revenue, profit, orders, customers and AOV — with no single dimension where another region overtakes it. In absolute terms the gap between the top and bottom region is meaningful: about 20.3M in revenue and about 3.1M in profit between North and South.

That said, customers are fairly evenly split across regions (roughly 25% each), so the gap is not extreme enough on its own to justify reallocating resources away from any single region. South's noticeably lower AOV likely reflects its category mix rather than a standalone regional issue — see Section 6.

6. Category's Leading Region

Each category paired with its single strongest-performing region. Figures are the category's overall totals (as in Section 3); the region column identifies where that category performs best, rather than a full per-region recalculation.

Category	Leading Region	Revenue	Profit	Orders	Cust.	AOV	Margin
Furniture	West	56,647,188	8,693,087	527	525	107,490	15%
Home Decor	North	57,233,222	8,556,847	515	513	111,132	15%
Clothing	West	55,053,908	8,445,750	511	510	107,738	15%
Books	South	54,932,643	8,076,273	528	524	104,039	15%
Electronics	East	52,587,884	8,042,134	472	472	111,415	15%
Toys	North	52,227,366	7,986,871	478	477	109,262	15%
Kitchen	East	54,227,902	7,879,573	528	527	102,704	15%
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Category	Leading Region	Revenue	Profit	Orders	Cust.	AOV	Margin
Beauty	South	50,803,410	7,213,436	460	460	110,442	14%
Groceries	North	47,883,103	7,075,333	470	468	101,879	15%

Insights & Comments

Beauty's leading region is South — the same region with the lowest AOV overall (Section 5). Since Beauty is also the weakest category by margin and order volume (Section 3), this pairing is consistent with South's lower average metrics being a category-mix effect rather than a standalone regional weakness. Confirming this with certainty would require a true category-within-region breakdown, which is a natural next step for this analysis rather than a conclusion drawn here.

7. Discount Impact Analysis

Objective: understand how discount size affects sales volume and profitability.

Discount	Orders	Total Sales	Avg Check	Total Profit	Avg Margin
0%	998	117,106,034	117,341	17,266,609	14.9%
5%	978	112,096,122	114,618	17,059,183	14.9%
10%	1,020	110,572,966	108,405	16,562,543	15.0%
15%	983	97,285,450	98,968	14,575,952	15.0%
20%	1,021	96,605,452	94,618	14,244,449	14.8%

Insights & Comments

The relationship between discount size and profitability is not linear. The highest discount tier (20%) does not translate into higher sales — it produces the lowest total sales and the lowest margin (14.8%) of all tiers, driven by a lower average check. Order counts are broadly similar across all discount levels, so the differentiator is check size and margin, not volume.

The 0% discount tier generates the highest profit overall. This pattern is worth exploring further at the category level in future analysis, but cannot be confirmed with the current dataset alone.

8. Payment Methods Analysis

Objective: explore payment methods used by customers and identify differences in order value and margin.

Payment Mode	Orders	Total Sales	Avg Check	Total Profit	Avg Margin
Net Banking	1,010	111,465,516	110,362	16,964,120	14.9%
COD	1,005	108,881,396	108,340	16,575,846	15.1%
Credit Card	994	106,027,145	106,667	15,658,639	14.9%
Debit Card	1,003	105,346,382	105,031	15,479,323	14.9%
UPI	988	101,945,586	103,184	15,030,807	14.9%

Insights & Comments

Performance is very consistent across all five payment methods, with no method standing out as materially stronger or weaker on sales, average check, or margin. This uniformity is itself a notable finding for a dataset of this size — it suggests payment method has little independent effect on customer behavior here, though a larger dataset would be needed to confirm this with confidence.

9. Top-10 Customers by Revenue

Customer Name	Orders	Total Sales	Avg Check	Total Profit	Avg Margin
Aaryahi Madan	3	650,152	216,717	107,156	14.8%

Customer Name	Orders	Total Sales	Avg Check	Total Profit	Avg Margin
Kimaya Vala	2	490,935	245,468	43,004	8.9%
Badal Rao	2	485,698	242,849	31,990	6.4%
Adira Dhingra	2	479,389	239,695	64,959	12.8%
Sumer Mann	2	469,937	234,969	99,925	21.5%
Nayantara Majumdar	2	459,764	229,882	58,949	11.2%
Prerak Shere	2	451,862	225,931	85,632	17.4%
Bhamini Sane	2	441,411	220,705	79,310	16.8%
Samarth Wagle	2	427,716	213,858	63,411	12.9%
Arhaan Basu	2	426,564	213,282	39,473	9.3%

Insights & Comments

Even the highest-value customers in this dataset are largely one-time or twice-only buyers — only one customer in the top 10 reordered a third time — confirming that this business is not built around repeat-customer retention, but around a high volume of large, one-off purchases.

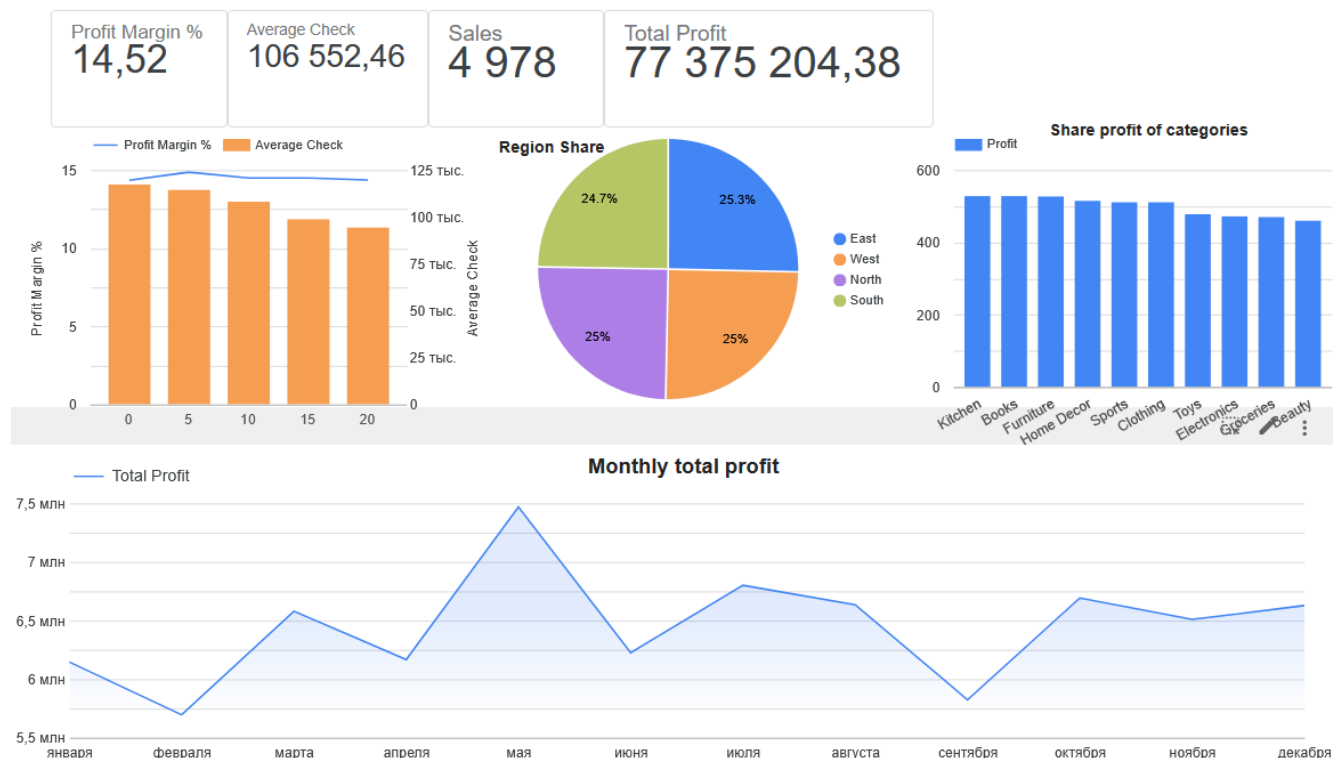
Margin varies significantly at the individual level, from about 6% to over 21%, which naturally balances out to the overall 15% average at scale. No specific action is indicated by this table on its own; it is included mainly as a sanity check on the aggregate figures reported elsewhere.

10. Visualization

Dashboard: <https://datastudio.google.com/reporting/828ef23e-b333-4791-b634-35ed35476d5c>

Insights & Comments

The dashboard consolidates the key metrics above into a single summary view of overall business performance across the analyzed period.



Looker Studio dashboard — summary view

11. Conclusions and Recommendations

The business demonstrates strong stability over the full two-year period: consistent margins, a healthy AOV, and no significant structural weaknesses across categories, regions, or payment channels.

The main finding worth flagging, however, is that stability has not been accompanied by growth: revenue and profit hold steady but do not trend upward across the two years. Whether this reflects a deliberate choice (a mature market, a conscious focus on margin over expansion) or an unaddressed growth ceiling cannot be determined from this dataset alone, but it is the single most actionable observation in this analysis.

Recommendations

- Investigate the root cause of the growth plateau before treating current stability as sufficient on its own.
- Compare the weaker end of the category range (Beauty) against the stronger end (Furniture, Home Decor, Electronics) to identify transferable factors.
- Given the consistently high AOV across categories and regions, prioritize cross-sell and bundling experiments over pure customer-acquisition spend.
- Treat the discount- and payment-method-level patterns as starting hypotheses rather than conclusions — a larger, multi-year dataset with additional dimensions (e.g. marketing spend, category-level discount rules) would be needed to confirm them with confidence.

Overall, this analysis is based on a single 5,000-row transaction table covering 2023–2025; broader conclusions about the business would require additional data (marketing activity, cost breakdowns, customer acquisition channels) that falls outside the scope of what was available here.